upon vast volcanic eminences on a geologically quiet Mars; when we discover the surface of Venus wiped clean only yesterday by floods of magma; when we find a world melted not by the heat of radioactive decay, as on Earth, but by gravitational tides exerted by nearby worlds; when we observe sulfur rather than silicate vulcanism; and when we begin to wonder, in the moons of the outer planets, whether we might be viewing water, ammonia, nitrogen, or methane vulcanism—then we are learning what else is possible.

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*The eruption of a nearby submarine volcano and the rapid construction of a new island in 197 B.C. are described by Strabo in the epigraph to this chapter.

*Even with its mountains and submarine trenches, our planet is astonishingly smooth. If the Earth were the size of a billiard ball, the largest protuberances would be less than a tenth of a millimeter in size—on the threshold of being too small to see or feel.

*The age of the Venus surface, as determined by *Magellan* radar imagery, puts an additional nail in the coffin of the thesis of Immanuel Velikovsky—who around 1950 proposed, to surprising media acclaim, that 3,500 years ago Jupiter spat out a giant "comet" which made several grazing collisions with the Earth, causing various events chronicled in the ancient books of many peoples (such as the Sun standing still on Joshua's command), and then transformed itself into the planet Venus. There are still people who take these notions seriously.

*Io's volcanos are also the copious source of electrically charged atoms such as oxygen and sulfur that populate a ghostly, doughnut-shaped tube of matter that surrounds Jupiter.